

OSI Reference Model

- What are the fundamental principles and characteristics of networking technologies like Ethernet, Token Ring, WLAN, and Bluetooth?

Purpose of OSI Model and What is it:

What is OSI Model: The OSI (Open Systems Interconnection) model is a conceptual reference framework that explains how data is transmitted from one computer to another over a network by dividing the communication process into seven distinct layers, each with a specific function.

What is the Purpose of OSI Model: Divide the complexity of the communication into several troubleshoot layers. To allow systems with different architectures and operating systems to communicate correctly. understand, design, and implement network communication. Organize how hardware, operating systems, and applications interact over a network. Provide a standard way of how communication of data takes place over the network.

TCP/IP, Hybrid, OSI model:

TCP/IP: A practical implementation of how networks work in the real world using 4 layered model.

Strength: Implemented widely in the real world, practical and robust. Simpler and more efficient than the OSI model.

Weakness: less clearly layered than the OSI model, harder to teach and to troubleshoot. No separate Session or Presentation layers.

Hybrid: A 5 layered model that combines the practicality of TCP/IP model with the clarity of the OSI model. Commonly used for teaching and industry discussions.

Strength: Practical for both learning and implementation, combines practicality with theoretical learning and implementation.

Weakness: not officially standardized, there are variations in many implementations so less standardization.

OSI: it has 7 layers and only is a conceptual implementation of how communication takes place on the network. Its used for understanding communication and troubleshooting.

Strength: Clear separation of layers, and responsibilities. Excellent for teaching and understanding how networks work. Provides a standard way for different network architectures from different vendors to communicate.

Weaknesses: more theoretical than practical, not implemented directly in real networks.

Layer Comparison in the 3 Models:

TCP/IP: Application Transport Internet link

Hybrid: Application Transport Internet (DataLink Physical) → we separated the link layer into 2 different Layers

OSI: Please Do Not Throw Sausage Pizza Away

(Application Presentation Session) → this was all inside the application layer. **Transport internet DataLink Physical.**

Aspects of 4 vs. 5 vs. 7 Layer Models:

Hybrid Model to distinguish duties of Physical vs. Data link Layer; in the TCP/IP model the link layer handles framing the data, error handling, and the actual transmission of bits through the physical media (radio waves, light signals, electric signals). In the hybrid model we separate this link layer into 2 layers data link layer and physical layer, this makes it easier to troubleshoot the area where an error has occurred, it makes it easier to understand.

OSI model separates tasks and duties in Application, Presentation and Session Layers. The other reference models combine all in Application Layer, which is mainly useful for computer networks.

OSI Model Structure (abstract description):

To memorize the layer names' we use this phrase down to up for the architecture:
"Please Don't Throw Sausage Pizza Away"

Application Layer: The Application layer takes a request from a network application, which receives user input, and then creates a protocol message such as HTTP, HTTPS, or SMTP to send over the network.

Presentation Layer: handles the formatting, encoding, decrypting of the data. It makes the data readable by the system

Session Layer: Establishes, manages, and terminates sessions, Controls communication sessions between systems

Transport Layer: Provides end-to-end communication, Handles reliability, flow control, and error recovery

Internet Layer: Handles logical addressing and routing, Determines the best path for data. Handles IP and other stuff. Adds the ip layering

Data Link Layer: handles framing and mac addressing.

Physical Layer: Transmits data across the physical media, Defines cables, signals, and transmission rates

Detailed Presentation Layer:

This layer makes sure that the data is in machine readable format by the other end device and all devices in between utf-8 unicode.

From a sender's perspective: making the data into readable format for other devices. Providing the data into bits/bytes.

Compressing the data.

Encoding the data from http message or xml or json into bits and byte readable by other machines and protocols.

Encrypting the data and making it secure using SSL/TLS protocols for website data encryption.

From a receiver's perspective: The layer takes data provided by session layer, decrypt it, decode it and decompress it. and make it possible for the application layer protocols to process it.

Detailed Session Layer:

The Session Layer is responsible for managing communication sessions between applications on two devices. Think of it as the layer that sets up, controls, and ends conversations between computers, ensuring data is properly synchronized during communication.

Management of session between end-user application processes; Initiation
Management Termination of communication (session).

Session Initiation:

- Starts a session between two devices or applications.
- Negotiates protocols, permissions, and parameters for the session.
- Example: Logging in to a remote server (SSH or ISDN connection).

Session Management:

- Maintains the session while communication continues.
- Controls the dialogue:
 - Half-duplex (one side talks at a time)
 - Full-duplex (both sides talk simultaneously)
- Tracks who can send data and when.
- Example: PPTP (Point-to-Point Tunneling Protocol) (PAP, PPTP, SCP)

Synchronization

- Adds checkpoints into a data stream so if a session is interrupted, communication can resume from the last checkpoint.
- decide on which layer to send what data when and whether it shall send using full- or half duplex
- Example: During file transfer, a checkpoint allows resuming instead of restarting from the beginning.

Detailed Physical Layer:

The Physical Layer is the lowest layer of the OSI model. Its main role is to physically transmit raw bits (0s and 1s) over the network medium. Think of it as the layer that turns the data into signals that can travel over wires, fiber, or wireless links.

Transmission of Bits on wired or wireless media raw data (bits, waves) does not care about the meaning or structure of this data.

Converts frames from Data Link Layer into signals suitable for the physical medium.

Device	Function
Repeater	Regenerates signals to extend transmission distance
Hub / Multiport Repeater	Forwards bits to all connected devices; no filtering, purely physical

Networking Technologies

Token Ring IEEE 802.5 (LAN):

It's a ring topology: In a ring topology, each device (terminal, PC, or workstation) is connected to exactly two other devices, forming a closed loop.

Data travels around the ring from one device to the next until it reaches the destination.

For a device to be able to transfer data to the next connected device it needs to acquire a small data frame called token so it can send data and when it receives data it shall send an acknowledgement to the device it received from.

Each terminal in a Token Ring network has an active network interface card (NIC) that actively participates in passing the token and regenerating the signal.

Token Ring networks use special cabling and connectors, often called MAU (Multistation Access Unit) connectors or IBM Data Connectors.

You cannot just use standard Ethernet cables without adapters.

IBM developed and promoted Token Ring, but as Ethernet became cheaper, faster, and simpler, Token Ring networks were gradually phased out.

Ethernet IEEE 802.3 (LAN):

It's the most widely used local area network

Robert Metcalfe and his team at Xerox PARC invented Ethernet in the early 1970s

In the OSI model ethernet works in the data link layer and physical layer.

Wired Ethernet with different electrical and fibre-optics media types (coaxial, Twisted Pair, Single Mode or Multi Mode Fibre-Optics).

Ethernet mostly uses baseband transmission, meaning digital signals are sent directly over the cable as a single signal. There is one rare historical exception where broadband transmission was used, but almost all Ethernet today is baseband; think of it like a one lane road.

Wireless LAN / Bluetooth (Problems Facing it):

Attenuation, fading over distance: similar to wired networks, but physical barriers (houses, walls and others) in addition

Hidden Terminal problem: devices are hidden due to physical obstacles (houses, walls, pillars, ...) → unintended interference

Multipath Propagation: Reflections of electromagnetic waves on physical obstacles, either fixed or moving

Moving terminals

Electromagnetic interference → Multiple use of frequency bands (eg. WiFi and BT)

→ Noise

Bluetooth Vs WiFi:

Bluetooth: Purpose & How It Works

- **Purpose:**
 - Provides short-range wireless communication between devices.
 - Designed for personal area networks (PANs) like headphones, keyboards, or smartwatches.
- **How It Works:**
 - Devices communicate over the 2.4 GHz ISM band using frequency hopping to reduce interference.
 - Connections are organized in piconets (one master, up to 7 active devices).
 - Uses protocols defined by the Bluetooth standard for pairing, authentication, and data exchange.

Wifi: Purpose & How It Works

Purpose:

- Provides wireless access to a local area network (LAN) for computers, smartphones, and IoT devices.
- Enables mobility, eliminating the need for physical cables.

How It Works:

- Devices connect to a wireless access point (AP).
- Data is transmitted using radio waves, typically in the 2.4 GHz or 5 GHz bands.
- Uses protocols like IEEE 802.11 (Wi-Fi) for framing, error detection, and authentication.
- Medium access is controlled using CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).